👩‍🏫 Staff Guide · Biology A · Unit 7 Famous Fossils & Deep Time (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Ms. Dunham's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 4 Famous Fossils: A4.1 Washington's famous fossils (stations), A4.2 Twine Time + Events in Evolution, A4.3 Building Bridges, Rock layers game (AMNH), UCMP webquests, Dinosaurs notebook check")	**Level 2 piece**
A4.1 Washington's famous fossils (8 stations)	Fossil Stations packet + notes part B
A4.2 Twine Time + Events in Evolution	Twine Time packet + notes part C
A4.3 Building Bridges + Rock layers game (AMNH)	Rock Layers & Building Bridges packet + notes parts A, C
UCMP webquests (fossil record, stories, T. rex)	Not modified; the slides cover fossil formation
Class slides	Deck: 20 slides, Words to Know first, hold-up-1-2-3 checks, recap
Her worksheets (grouped into lessons)	Guided notes: 17 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	10 picture cards (cut apart)
—	2 games on the class site: matching + step-by-step order

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Picture anchors: 🎂 rock layers = layer cake, 🦣 state fossil, 🕰️ people came last
- Level 1 uses order (first/last) instead of numbers of years

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Fossils quiz stands in for the Dinosaurs notebook check content
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Ms. Dunham".

🚧 Under construction (check with Ms. Dunham)

- **Twine Time dates**: Her "WA fossils_TwineTime" cards give dates that do not match her stations sheet (for example Stone Rose 10 vs 50 million years, Blue Lake Rhino 66 million). Our games use order, plus the dates from her stations sheet. Please check which dates she wants.

6. Photo credits (open licenses)

Columbian mammoth.JPG — WolfmanSF, CC BY-SA 3.0, Wikimedia Commons · Big Bend Santa Elena Canyon 2006.JPG — Leaflet, Public domain, Wikimedia Commons · Ginkgo Petrified Forest State Park near Vantage along the Columbia River 2.jpg — Thayne Tuason, CC BY-SA 4.0, Wikimedia Commons · Fossil leaf structure detail, Ginkgo biloba leaf 01 (cropped).jpg — Kevmin, CC BY-SA 4.0, Wikimedia Commons

👩‍🏫 Staff Guide · Washington Fossil Stations (Level 2)

Biology A · Unit 7 · goes with the class “A4.1 Washington's Famous Fossils” (5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Station card**: A4.1 Washington's Famous Fossils: 8 stations (mammoth, first dinosaur, Clovis, Blue Lake Rhino, Seatac Sloth, Stone Rose, Ginkgo forest, Diatryma) with models and fossil casts. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Flat teeth or sharp teeth?”
- “Which rhino fits the hole?”

3. Modifications

- 6 of 8 stations
- Questions with an idea bank

4. What to grade (5 pts)

- Same 5 points: 6 stations + reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

6. Photo credits (open licenses)

Columbian mammoth.JPG — WolfmanSF, CC BY-SA 3.0, Wikimedia Commons

👩🏾‍🏫 Staff Guide · Rock Layers & Building Bridges (Level 2)

Biology A · Unit 7 · goes with the class “A4.3 Building Bridges + Rock layers game” (5 + 5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Activity card**: Rock layers game (AMNH) + A4.3 Building Bridges: read fossil columns in rock layers and trace mammals across the Central American land bridge. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Which layer was made first?”
- “Show me how the armadillo walked north.”

3. Modifications

- 4 layers; map pre-labeled
- Whiteboard presentation optional

4. What to grade (5 + 5 pts)

- Same points: layers + map + reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

6. Photo credits (open licenses)

Big Bend Santa Elena Canyon 2006.JPG — Leaflet, Public domain, Wikimedia Commons

👩‍🏫 Staff Guide · Twine Time Timeline (Level 2)

Biology A · Unit 7 · goes with the class “A4.2 Twine Time + Events in Evolution” (5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Timeline card**: A4.2 Twine Time: sort event cards (biotic/abiotic), predict their order, then place them on a 5-meter twine timeline of Earth's history. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “What came first: fish or dinosaurs?”
- “Where do people go?”

3. Modifications

- 8 cards with dates printed
- Scale with a word bank

4. What to grade (5 pts)

- Same 5 points: order + scale + reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

🚧 Under construction (check with Ms. Dunham)

- **This packet**: Her Twine Time cards and Events in Evolution sheet are pictures only. We made our own 8 event cards with standard dates. Please check they match her events.

🦕 Famous fossils notes (A4.1-A4.3)

Biology A · Unit 7 Famous Fossils & Deep Time · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Fossils and rock layers (A4.1, Rock layers game)

1. A fossil forms best when an organism is buried _____.
2. Over time, _____ replace bone or wood.
3. Older rock layers are _____.
4. Number how a fossil forms (1-4):

_____ buried in fine sediment

_____ erosion uncovers it

_____ organism dies

_____ minerals replace hard parts

Word bank: **deeper** **minerals** **quickly**

B. Washington's famous fossils (A4.1 stations)

5. Washington's state fossil is the Columbian _____.
6. Petrified wood at Vantage is from _____ trees.
7. Match the fossil to the clue:

Blue Lake Rhino	stone tools of early people
Seatac Sloth	giant bird footprints
Clovis points	mold in lava rock
Diatryma	giant plant eater

Word bank: **ginkgo** **mammoth**

C. Deep time and land bridges (A4.2, A4.3)

8. Earth is about 4.6 _____ years old.
9. A _____ bridge connects two continents.
10. Animals found in only one region are _____.
11. Circle the 2 animals that came north over the land bridge:

armadillo · opossum · mammoth · orca

Word bank: **billion** **endemic** **land**

Level 2 version of the gen ed “A4.1 Washington's Famous Fossils + stations, A4.2 Twine Time, A4.3 Building Bridges, Rock layers game.”

📖 Famous Fossils & Deep Time: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.

🦴

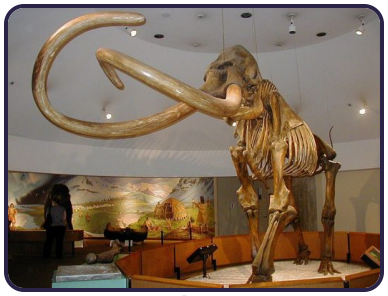
Fossil

The remains or traces of a living thing from long ago, kept in rock.



Rock layers

Layers of rock (strata). Older layers are at the bottom.



Mammoth

The Columbian mammoth is Washington's state fossil.

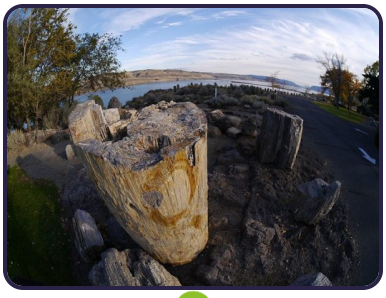
⌛

Extinct

When every member of a species has died.

Deep time

Earth is about 4.6 billion years old. Life began about 3.5 billion years ago.



Petrified wood

Minerals slowly replace wood, like at Ginkgo Petrified Forest in Vantage.



Imprint

A fossil shape pressed into mud that turned to rock, like a leaf print.

Land bridge

Land that connects continents so animals can move between them.

Endemic

Found naturally in only one region.

Paleontologist

A scientist who studies fossils to learn about past life.

🦣 Washington Fossil Stations

Biology A · Unit 7 · Level 2



Name: _____

A4.1 Washington's Famous Fossils: 8 stations (mammoth, first dinosaur, Clovis, Blue Lake Rhino, Seatac Sloth, Stone Rose, Ginkgo forest, Diatryma) with models and fossil casts.

🧺 Stations we do

- ☐ 🦣 **1** Columbian mammoth: teeth + models
- ☐ 🦏 **4** Blue Lake Rhino: fossil mold
- ☐ 🍂 **6** Stone Rose: leaf fossils + clay imprint
- ☐ 🌳 **7** Ginkgo petrified forest

🌱 Steps: check each box when you finish

1. 🦣 **Mammoth**: compare the teeth. Which eats grass best? Compare the toy and skeleton models. ☐
2. 🦖 **First dinosaur**: why are dinosaur fossils rare in Washington? ☐
3. 🦏 **Blue Lake Rhino**: what conditions make a fossil? Which rhino fits the mold? ☐
4. 🦥 **Seatac Sloth**: plant eater or meat eater? Use the claw and tooth as evidence. ☐
5. 🍂 **Stone Rose**: press a leaf in clay. Compare it to the 50-million-year-old leaf. ☐
6. 🌳 **Ginkgo forest**: how does wood turn to stone? ☐

🦣 Mammoth models

	Toy model	Wooden skeleton
Good model because…		
Not like a real mammoth because…		

🌉 Rock Layers & Building Bridges

Biology A · Unit 7 · Level 2



Name: _____

Rock layers game (AMNH) + A4.3 Building Bridges: read fossil columns in rock layers and trace mammals across the Central American land bridge.

🧺 Activity stuff

- ☐ 🪨 **4 strips** paper rock layers (bottom = oldest)
- ☐ 🦴 **fossil stickers** or drawings
- ☐ 🗺️ **1 map** North + South America
- ☐ 🖍️ **highlighters** trace who crossed

🌱 Steps: check each box when you finish

1. 🪨 **Stack** 4 layers in order. The first layer laid down goes at the bottom. ☐
2. 🦴 **Place** the fossils. A species is a column through the layers it lived in. ☐
3. ⏳ **Find** where a column ends: that species went extinct. ☐
4. 🗺️ **Compare** the two maps: before and after the land bridge. ☐
5. 🌉 **Trace** armadillo and opossum moving north. ☐
6. 📍 **Decide**: which animals were endemic to South America? ☐

📋 Land bridge

Animals that moved north: _____ Animals that moved south: _____

Ideas: **armadillo** **opossum** **deer** **cats**

🕰️ Twine Time Timeline

Biology A · Unit 7 · Level 2



Name: _____

A4.2 Twine Time: sort event cards (biotic/abiotic), predict their order, then place them on a 5-meter twine timeline of Earth's history.

🧺 Event cards (our own)

- ☐ 🌍 **Earth forms** 4.6 billion years ago
- ☐ 🦠 **First life (algae, bacteria)** 3.5 billion
- ☐ 🐟 **First fish** about 500 million
- ☐ 🦖 **First dinosaurs** about 230 million
- ☐ 🌸 **First flowers** about 130 million
- ☐ ☄️ **Dinosaurs go extinct** 66 million
- ☐ 🦣 **Mammoths** Ice Age
- ☐ 🧍 **Humans** about 300,000

🌱 Steps: check each box when you finish

1. ✂️ **Cut** the 8 event cards. ☐
2. 🗂️ **Sort**: living (biotic) or non-living (abiotic) events. ☐
3. 🔢 **Predict** the order from oldest to newest. ☐
4. 📏 **Scale**: 1 m = 1 billion years, so 1 cm = 10 million years. ☐
5. 📎 **Place** each card on the twine at its spot. ☐
6. 🧠 **Discuss**: where are humans? What surprised you? ☐

📏 Scale

1 m = _____ years · 1 cm = _____ years

Humans are _____ cm from the end of the twine.

Word bank: **1 billion** **10 million** **0.03**

🚧 **Under construction:** Her Twine Time cards and Events in Evolution sheet are pictures only. We made our own 8 event cards with standard dates. Please check they match her events. *Check with Ms. Dunham.*

🦣 Washington Fossil Stations · Prep Sheet

Biology A · Unit 7 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
📖 **Reader**	Reads the station sign out loud.	
🔍 **Looker**	Holds and compares the fossil and the model.	
📝 **Recorder**	Circles or writes the answers.	
⏱️ **Timer**	Says “switch!” when time is up.	

🧰 Tools and supplies: check when you have it

- ☐ 🧱 **Clay** make a leaf imprint
- ☐ 🍃 **Real leaf** press into clay
- ☐ 🦴 **Fossil casts and models** look and compare, handle gently
- ☐ 📋 **This card** one box per station

⚠️ Safety and manners focus

- 🦴 Fossils and casts break easily: two hands, keep them on the table.
- 🧼 Wash hands after the clay.
- 🚶 Walk between stations.

🌉 Rock Layers & Building Bridges · Prep Sheet

Biology A · Unit 7 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🪨 **Stacker**	Stacks the rock layers from the bottom up.	
🦴 **Fossil placer**	Puts each fossil in its layer.	
🗺️ **Map tracer**	Traces animals across the land bridge.	
📝 **Recorder**	Writes oldest and youngest.	

🧰 Tools and supplies: check when you have it

- ☐ 🪨 **Layer strips** stack from the bottom up
- ☐ 🗺️ **Map** land bridge
- ☐ 🖍️ **Highlighter** trace animal paths
- ☐ 💻 **AMNH game (optional)** Layers of Time, staff helps

⚠️ Safety and manners focus

✂️ Scissors points down.

💻 Laptops: two hands, on the table.

🕰️ Twine Time Timeline · Prep Sheet

Biology A · Unit 7 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
✂️ **Card cutter**	Cuts the event cards apart.	
🔢 **Orderer**	Puts the cards in order.	
📏 **Measurer**	Finds the spot on the twine (1 m = 1 billion years).	
📎 **Hanger**	Clips the cards on the twine.	

🧰 Tools and supplies: check when you have it

- ☐ 🧵 **Twine** about 5 m = 5 billion years
- ☐ 📎 **Clips or tape** hang the cards
- ☐ 📏 **Meter stick** 1 m = 1 billion years
- ☐ ✂️ **Scissors** cut the cards

⚠️ Safety and manners focus

✂️ Scissors: walk with points down.

🧵 Keep the twine low and out of walkways.

🦣 Washington Fossil Stations · Reflection

Biology A · Unit 7 · Level 2



Name: _____

📸 My photos / proof

☐ Clay imprint ☐ Station card

✅ Skills I used today (check all you did)

☐ 🔍 I compared models with real fossils

☐ 🍂 I made a clay imprint

☐ 🦷 I used teeth/claws as evidence

⭐ Which Washington fossil was most interesting? Why?

1. _____
2. _____

Ideas: **mammoth** **rhino** **sloth** **ginkgo**

🦴 Why are whole skeletons rare?

1. _____
2. _____

Ideas: **eaten** **not buried fast** **broken**

🌉 Rock Layers & Building Bridges · Reflection

Biology A · Unit 7 · Level 2



Name: _____

📸 My photos / proof

☐ Layer stack ☐ Map

✅ Skills I used today (check all you did)

☐ 🪨 I ordered layers and fossils

☐ 🗺️ I traced migration

☐ 🤝 I did my role

⭐ How do you know a deeper layer is older?

1. _____
2. _____

Ideas: **laid down first** **stacked** **bottom**

🌉 What happened when the land bridge formed?

1. _____
2. _____

Ideas: **animals moved** **north** **south**

🕰️ Twine Time Timeline · Reflection

Biology A · Unit 7 · Level 2



Name: _____

📸 My photos / proof

☐ Timeline photo ☐ Scale box

✅ Skills I used today (check all you did)

☐ 🔢 I predicted and revised the order

☐ 📏 I used the scale

☐ 🤝 I did my role

⭐ What surprised you about the timeline?

1. _____
2. _____

Ideas: **humans so late** **life so early** **dinosaurs**

🦠 Why are different life forms found at different times?

1. _____
2. _____

Ideas: **evolution** **Earth changed** **extinction**

✅ Famous Fossils & Deep Time Quiz

Biology A · Unit 7 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🟫 Fossils form best when an organism is…

A. Left in the open | B. Buried quickly in fine sediment | C. Burned

2. 🪨 Petrified wood forms when…

A. Wood is painted | B. Wood freezes | C. Minerals slowly replace the wood

3. ⬇️ At one site, the oldest rock layer is…

A. At the bottom | B. At the top | C. In the middle

4. ⏳ When a species goes extinct, its fossil column…

A. Gets wider | B. Moves to the bottom | C. Ends

5. 🦣 Washington's state fossil is the…

A. Columbian mammoth | B. T. rex | C. Blue Lake Rhino

6. 🦥 The Seatac Sloth was a…

A. Meat eater | B. Plant eater | C. Fish

7. 🗿 Clovis points are…

A. Dinosaur teeth | B. Stone tools of early people | C. Shells

8. 🌍 Earth is about…

A. 4.6 billion years old | B. 6,000 years old | C. 1 million years old

9. 🌉 A land bridge lets animals…

A. Fly | B. Turn into fossils | C. Move between continents

10. 📍 Endemic means…

A. Found in only one region | B. Found everywhere | C. Extinct

🦕 Famous fossils notes (A4.1-A4.3) · ANSWER KEY

Biology A · Unit 7 Famous Fossils & Deep Time · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Fossils and rock layers (A4.1, Rock layers game)

1. A fossil forms best when an organism is buried **quickly**.
2. Over time, **minerals** replace bone or wood.
3. Older rock layers are **deeper**.
4. Number how a fossil forms (1-4):

2 buried in fine sediment
4 erosion uncovers it
1 organism dies
3 minerals replace hard parts

Word bank: deeper · minerals · quickly

B. Washington's famous fossils (A4.1 stations)

5. Washington's state fossil is the Columbian **mammoth**.
6. Petrified wood at Vantage is from **ginkgo** trees.
7. Match the fossil to the clue:

(Key: Blue Lake Rhino → mold in lava rock · Seatac Sloth → giant plant eater · Clovis points → stone tools of early people · Diatryma → giant bird footprints)

Blue Lake Rhino	stone tools of early people
Seatac Sloth	giant bird footprints
Clovis points	mold in lava rock
Diatryma	giant plant eater

Word bank: ginkgo · mammoth

C. Deep time and land bridges (A4.2, A4.3)

8. Earth is about 4.6 **billion** years old.

9. A **land** bridge connects two continents.

10. Animals found in only one region are **endemic**.

11. Circle the 2 animals that came north over the land bridge:

armadillo **opossum** mammoth orca

Word bank: **billion** **endemic** **land**

Level 2 version of the gen ed "A4.1 Washington's Famous Fossils + stations, A4.2 Twine Time, A4.3 Building Bridges, Rock layers game."

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

✅ Famous Fossils & Deep Time Quiz · ANSWER KEY

Biology A · Unit 7 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🟫 Fossils form best when an organism is… A4.1 stations

- A. Left in the open
- **B. Buried quickly in fine sediment**
- C. Burned

2. 🪨 Petrified wood forms when… A4.1 stations

- A. Wood is painted
- B. Wood freezes
- **C. Minerals slowly replace the wood**

3. ⬇️ At one site, the oldest rock layer is… Rock layers game

- **A. At the bottom**
- B. At the top
- C. In the middle

4. ⏳ When a species goes extinct, its fossil column… Rock layers game

- A. Gets wider
- B. Moves to the bottom
- **C. Ends**

5. 🦣 Washington's state fossil is the… A4.1 stations

- **A. Columbian mammoth**
- B. T. rex
- C. Blue Lake Rhino

6. 🦥 The Seatac Sloth was a… A4.1 stations

- A. Meat eater
- **B. Plant eater**
- C. Fish

7. 🗿 Clovis points are… A4.1 stations

- A. Dinosaur teeth
- **B. Stone tools of early people**
- C. Shells

8. 🌍 Earth is about… A4.2 Twine Time

- **A. 4.6 billion years old**
- B. 6,000 years old
- C. 1 million years old

9. 🌉 A land bridge lets animals… A4.3 Building Bridges

- A. Fly
- B. Turn into fossils
- **C. Move between continents**

10. 📍 Endemic means… A4.3 Building Bridges

- **A. Found in only one region**
- B. Found everywhere
- C. Extinct

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed "A4.1 WA famous fossils + stations, A4.2 Twine Time, A4.3 Building Bridges, Rock layers game". Score: __ / 10.

🔑 *Key written by our team from her A4.1-A4.3 sheets, please check. There is no Fossils test in the export; this is a "From notes" unit quiz.*